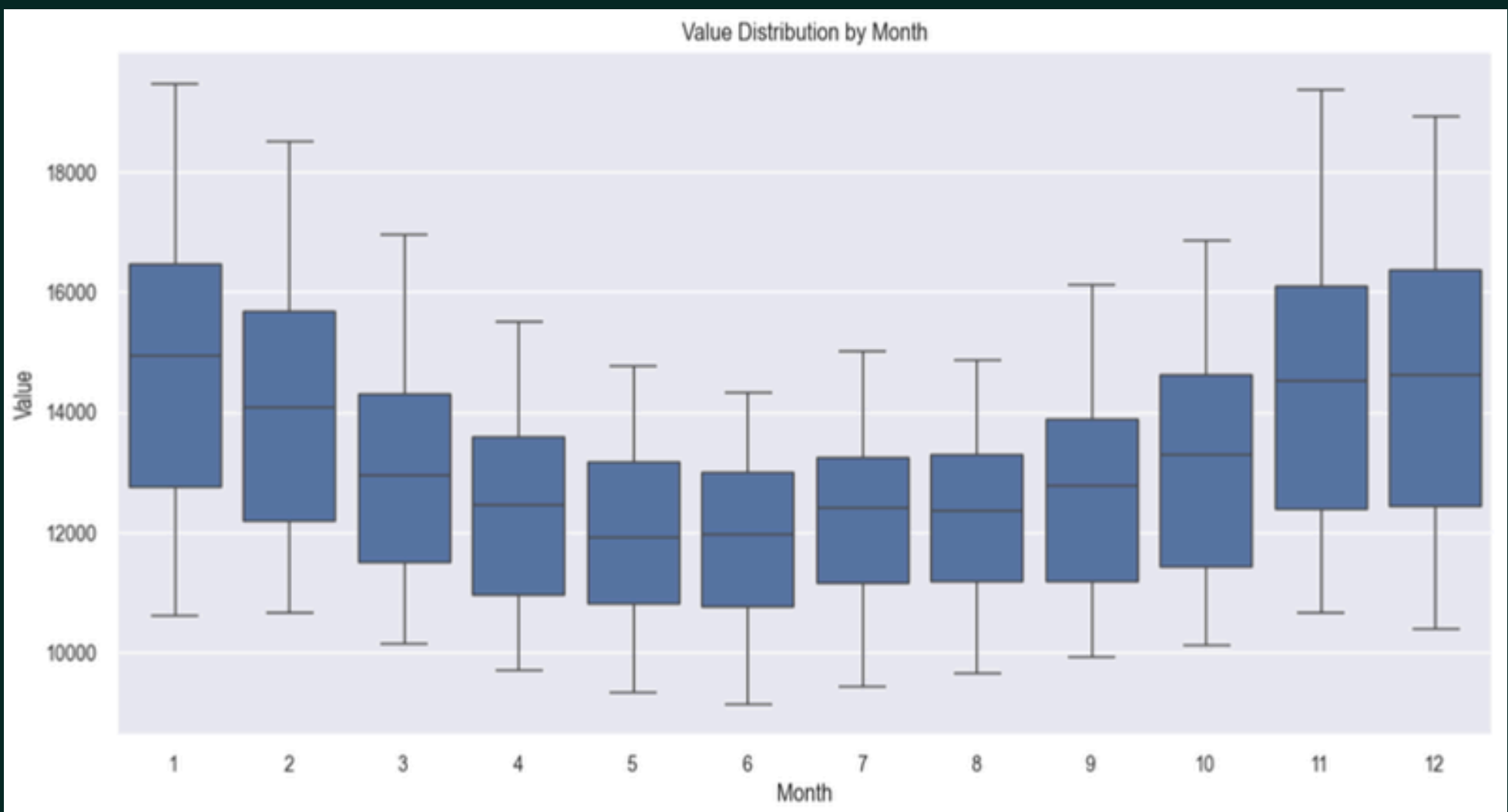
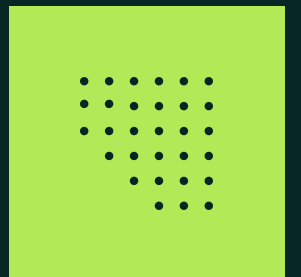




ENTSO-E Data



 **Mohamed Tawhid**
Energy Analyst | Power System Specialist

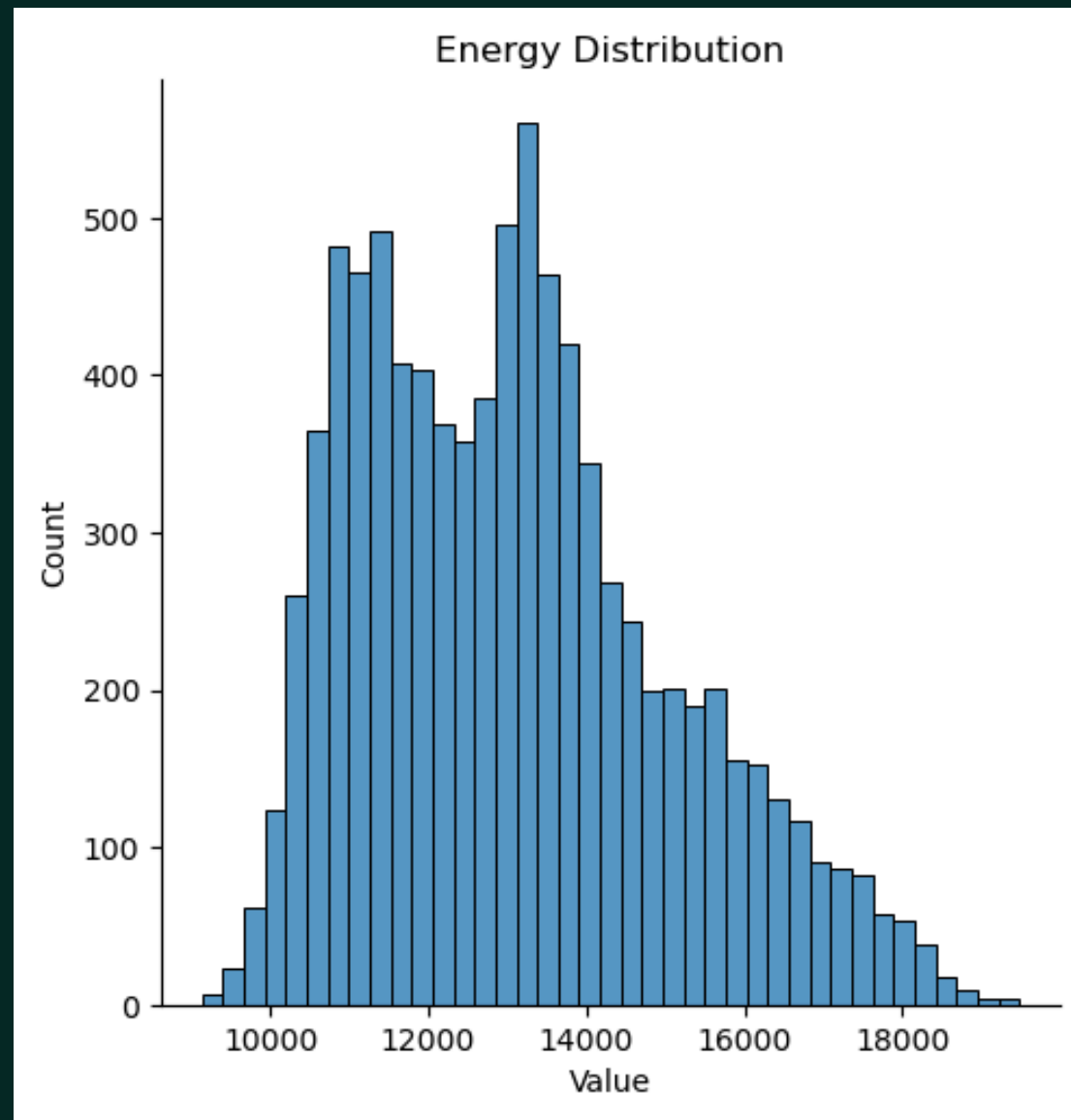
Energy Consumption Analysis Case Study

Decoding the Netherlands' Energy Demand
A Data-Driven Analysis





What Drives a Nation's Electricity Consumption?



Energy Distribution



Objective: To analyze the Netherlands' 2024 hourly load data to identify patterns, predict peaks, and provide actionable strategies for grid optimization.

Dataset: ENTSO-e Transparency Platform

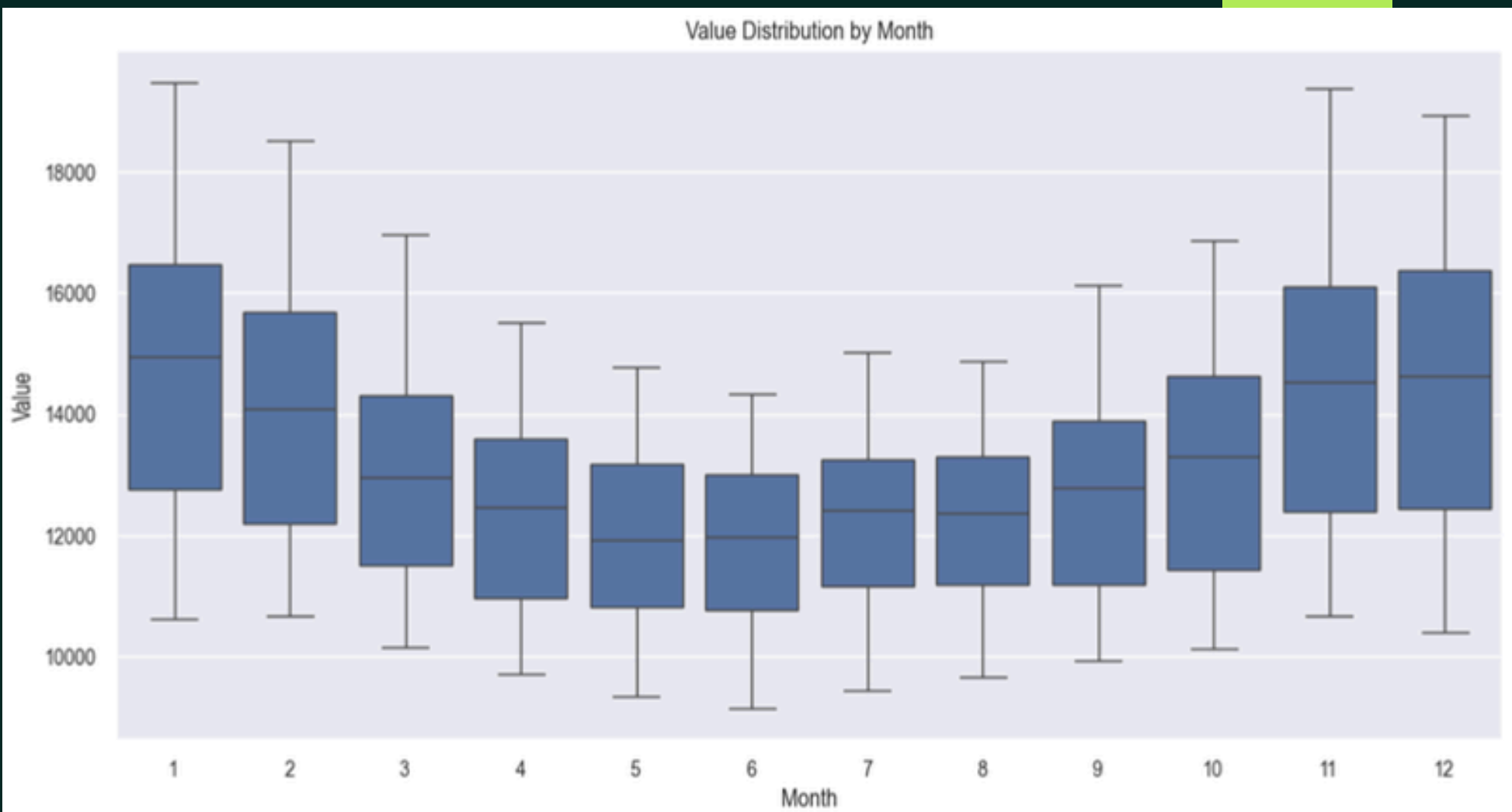
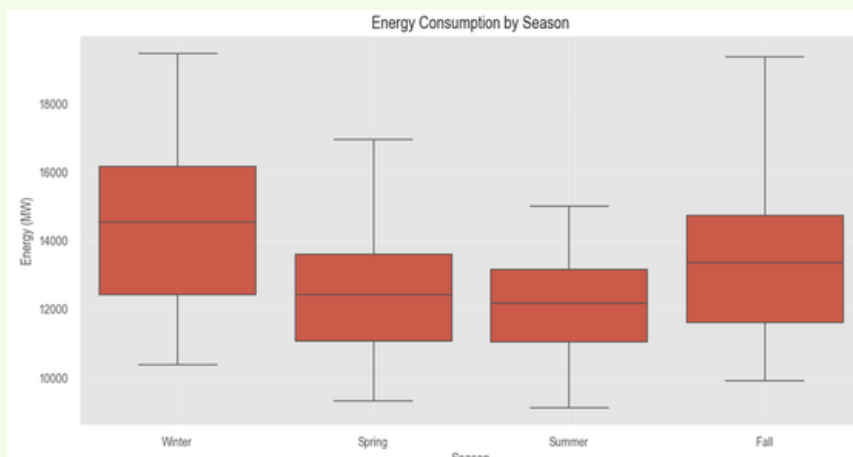
Scope: 8,784 hours | Full Calendar Year 2024 | Country Code: NL

••• CRITICAL FINDINGS

Demand Heavily Depends on the Season

The boxplot clearly reveals a seasonal shape – higher medians and spreads in winter (Dec–Feb), lower in summer (Jun–Aug).

Key Takeaway: Capacity planning must be seasonal.



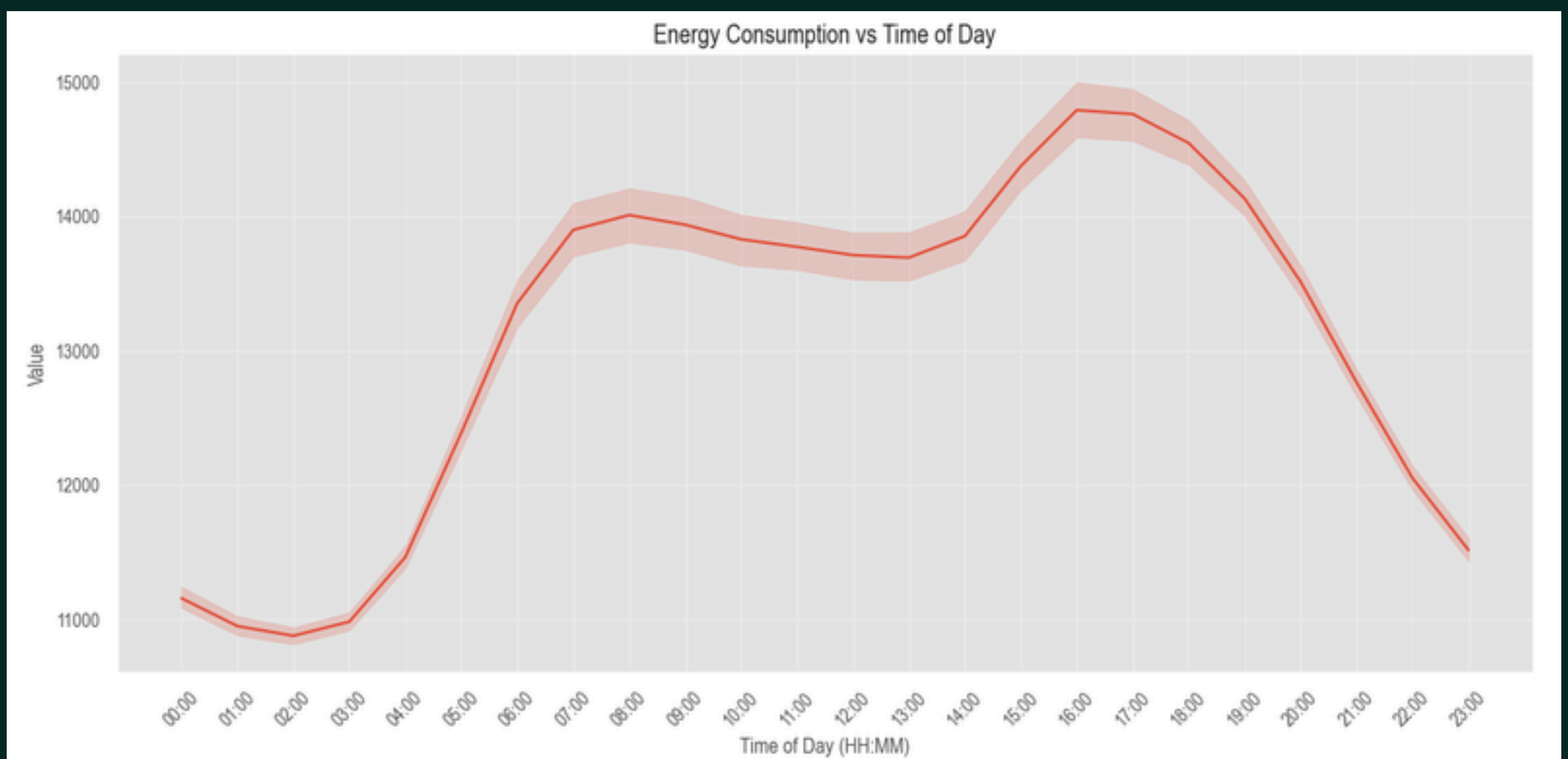
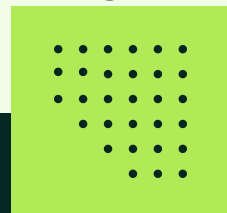
Value Distribution by Month

• • • The Daily "Duck Curve"

The Grid's Most Critical Hours: 6 PM - 8 PM

- **Overnight Low:** Load hits its minimum around 4-5 AM (~10,500 MW).
- **Morning Rise:** Sharp increase starting at 6 AM as the country wakes up.
- **Evening Peak:** The highest demand occurs between 6-8 PM.

Key Takeaway: This predictable pattern is critical for scheduling energy generation and planning demand-response programs.



Energy Consumption vs Time of Day

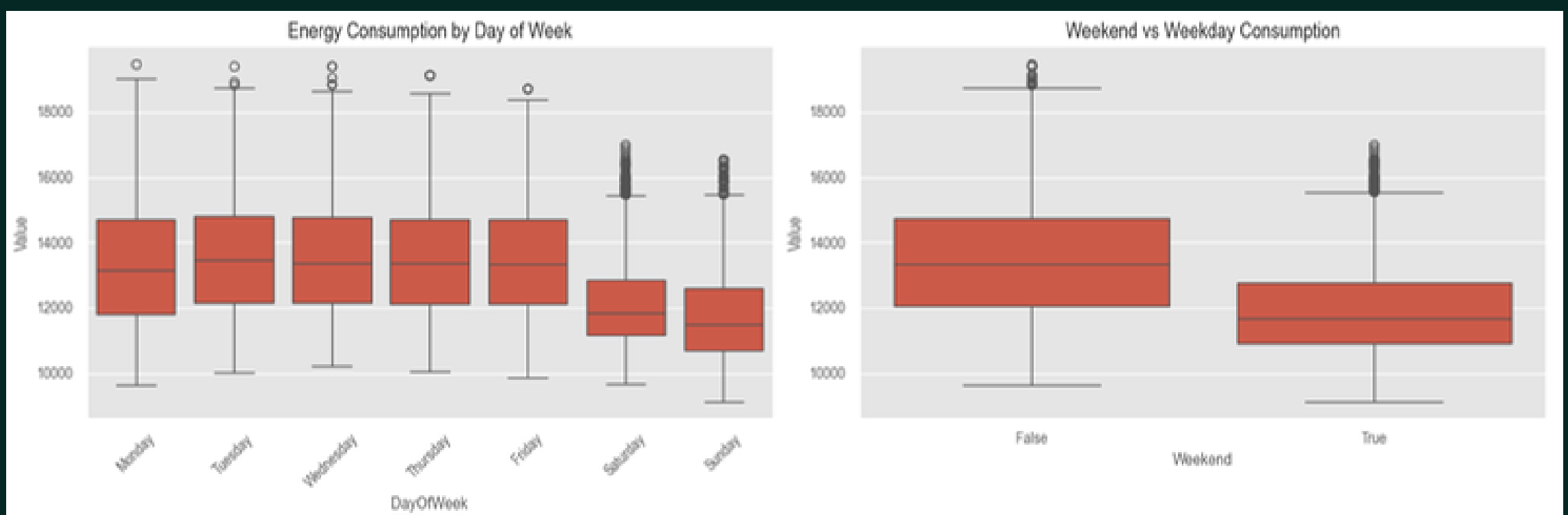
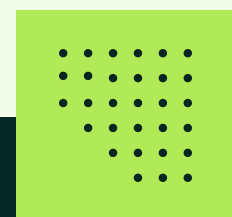


● ● ● Weekday vs. Weekend: The Commercial Impact

Weekends Tell a Different Story

A clear and consistent drop in energy consumption during weekends highlights the massive impact of commercial and industrial activity.

Key Takeaway: Different operational strategies are needed for weekdays and weekends.



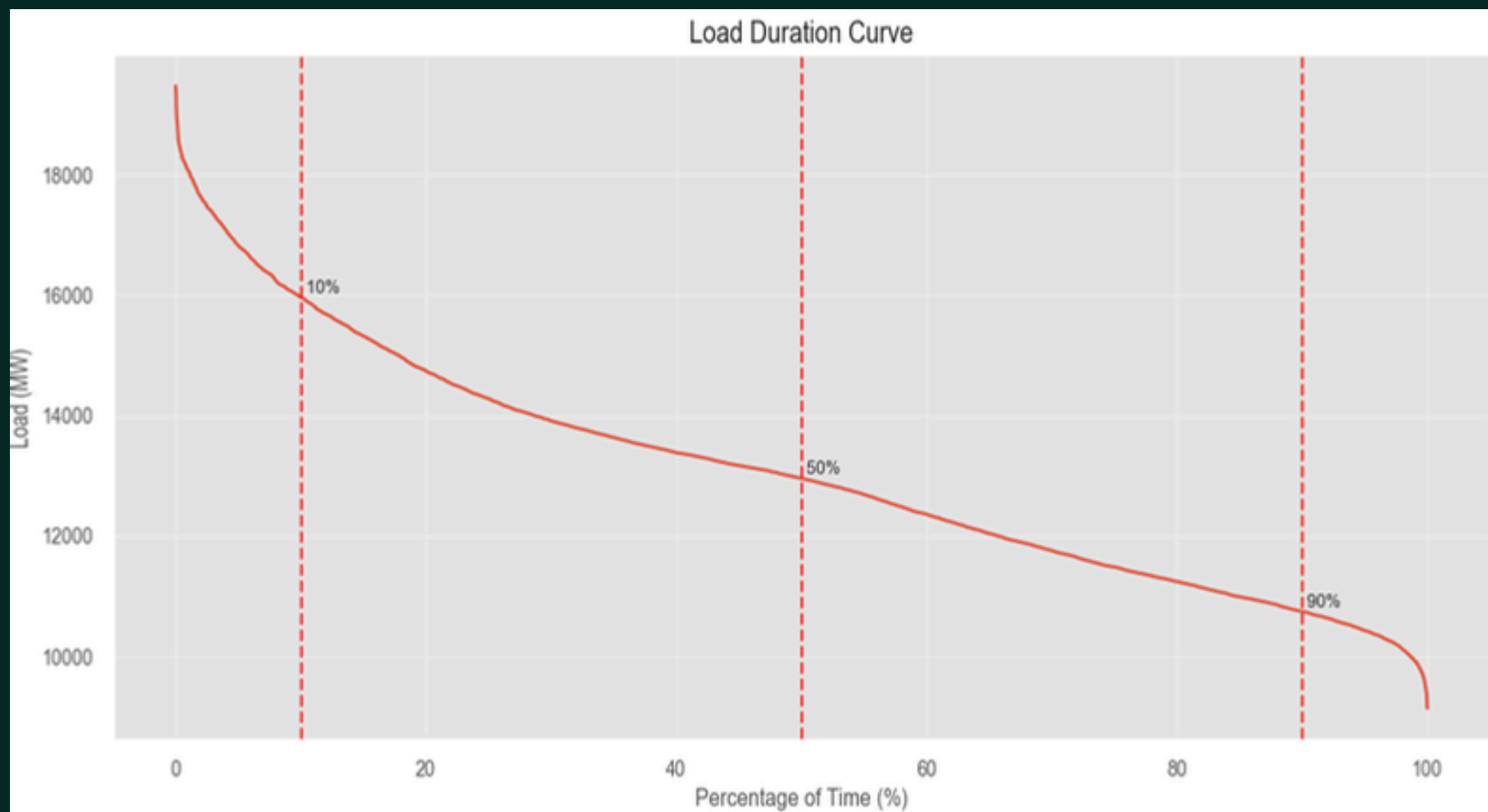
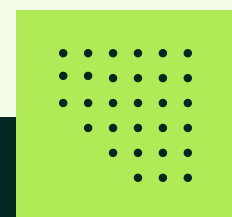
● ● ● The Load Duration Curve: A Planner's Best Tool

How Often Do We See Extreme Demand?

The Load Duration Curve sorts demand from highest to lowest, showing how frequently the grid is under stress.

- Top 10% of time: Load > 16,000 MW
- Bottom 10% of time: Load < 10,750 MW

Key Takeaway: Focus resources on the top 10% of demand hours.



Load Duration Curve



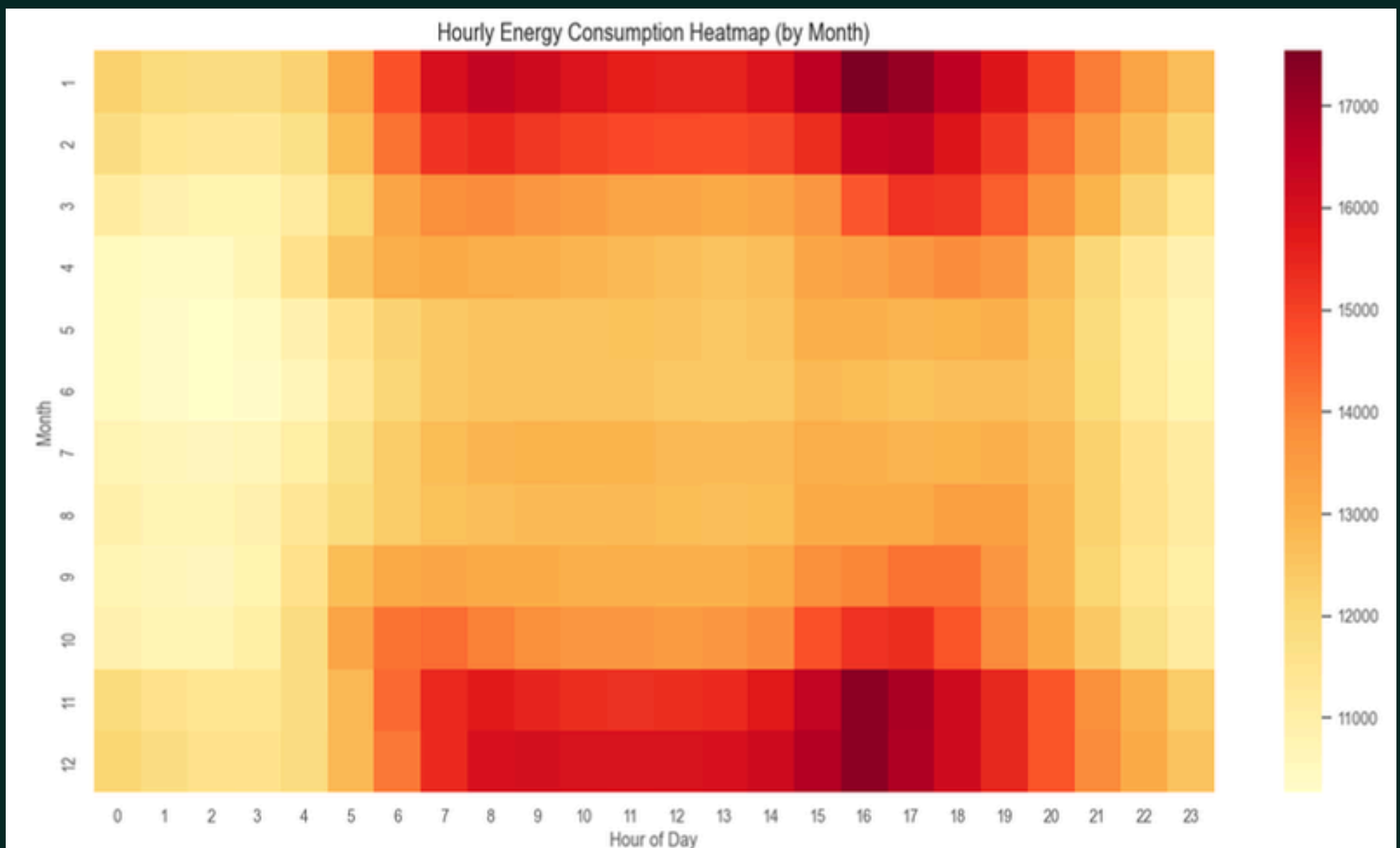
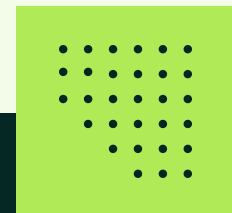
● ● ● The Hour-by-Hour, Month-by-Month Story

The Entire Year at a Glance

This heatmap visualizes the synergy between time of day and month of the year. Darker reds indicate higher consumption, clearly showing winter mornings and evenings are the most intense periods.

Key Takeaways:

The diurnal pattern flattens in summer and intensifies in winter. Heatmap confirms winter evenings are “hot”, summer nights are “cool”.



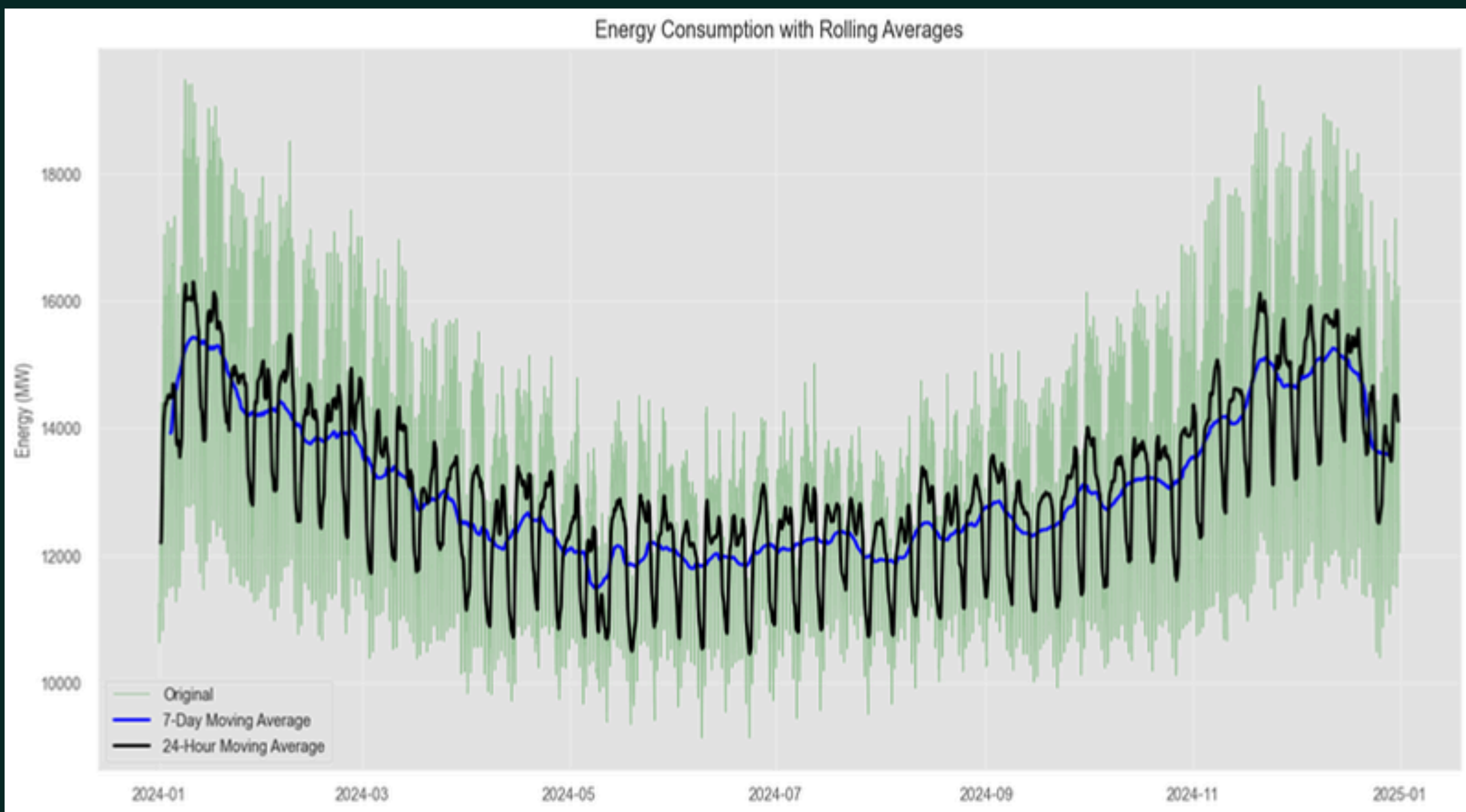
Strategic Recommendations

- **Grid Maintenance:** Schedule critical maintenance in Spring (April/May) or Autumn (Sept/Oct) when loads are moderate, avoiding winter stress.
- **Demand Response:** Target evening peaks (6-8 PM) for demand-side management. Incentivize shifting industrial loads to overnight hours (4-6 AM).
- **Forecasting Features:** For predictive models, prioritize [IsWeekend] and [HourOfDay] as key features—they are the strongest predictors of load.

 Mohamed Tawhid

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Call to Action



Energy Consumption with Rolling Averages